
Universal Plastic Shredder V2.0

Electrical Control & Power System

PORTLAND STATE UNIVERSITY
MASEEH COLLEGE OF ENGINEERING & COMPUTER SCIENCE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Final Report — Rough Draft

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[NOTE: This is a rough draft. Sections marked TODO, NOTE, or QUESTION indicate areas where we need to expand content, gather data, or get advisor input before the final submission. Bullet points are placeholders for fuller prose in the final version.]

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1 Executive Summary

- Ichor Systems generates significant plastic waste (PVC, polypropylene, LDPE) during semiconductor equipment manufacturing, currently sent to landfill.
- Commercial industrial shredders cost \$10,000+ and consumer units cannot handle the thick rigid sheets (0.25–0.5 in).
- Our project designs the electrical control and power system for a custom industrial-grade plastic shredder, capable of shredding into 1-inch pieces.
- Key elements: VFD-controlled 3 HP motor at 70–90 RPM output, two-hand deadman control, hardwired E-stop, jam-detection auto-recovery with 3-strike lockout, NEMA-enclosed PLC and HMI.
- Total budget: \$3,000 (EE portion approximately \$1,000).

[QUESTION: Should the executive summary include high-level results / status (e.g., “shredder validated to X HP load with Y throughput rate”) once we complete testing? Currently we have signal-chain validation but not full mechanical load testing.]

2 Introduction

2.1 Background and Motivation

- Ichor Systems is a Portland-area semiconductor equipment manufacturer.
- Their manufacturing processes produce substantial volumes of rigid plastic sheet waste.
- Existing disposal route: landfill – environmentally and economically suboptimal.
- A previous 2023–2024 PSU capstone team built a 2 HP household shredder for the EPL. This project scales to industrial-grade.

2.2 Project Scope

- EE team responsibility: motor control, safety circuits, PLC logic, HMI, power distribution.
- ME team (separate) handles mechanical frame, blade assembly, gearbox.

- Integration plan: mechanical structure delivered with motor mount; EE team commissions controls.

[TODO: Add a one-paragraph background on dynamic recycling / circular economy framing for the introduction.]

3 Requirements

3.1 Must-Have

- Shred 0.25–0.5 in plastic sheet into approximately 1-inch pieces.
- Controllable VFD supporting 3–5 HP motor at 70–90 RPM output shaft speed.
- Follow NFPA 70 (NEC) electrical design guidance.
- Include emergency stop, overcurrent/overload protection, at least one safety interlock.
- Include an HMI for operator control and status display.
- Stay within \$3,000 total project budget.

3.2 Should-Have

- Clear status and fault indicator lights.
- Follow NEMA enclosure guidelines.
- Clearly labeled controls.

3.3 May-Have (if time/budget allow)

- Wireless monitoring (read-only).
- Data logging of operating hours and faults.
- Additional safety sensing (capacitive touch, thermal).

4 System Architecture

4.1 Hardware Subsystems

- **AC power input:** 120 VAC mains → main disconnect → branch breakers → VFD and 24V PSU.

- **VFD:** Huanyang HY02D211B-T (2.2 kW, 110V single-phase input, 220V 3-phase output boost-type). Controlled via FOR/REV/RST digital inputs.
- **Motor:** bench/test unit is a GE 5KE182BC205B (3 HP, 230 V low-voltage connection, 1750 RPM, FLA 7.6 A); its nameplate drives the VFD motor parameters. **[TODO: Final production motor selection is TBD (3–5 HP per requirements).]**
- **Brake resistor:** 75 Ω , 500 W, connected to VFD's P/Pr terminals to handle regenerative energy during deceleration and reversal.
- **PLC:** AutomationDirect H2-DM1E (Do-more CPU) on DL205 base, D2-08ND3 discrete input module.
- **HMI:** AutomationDirect C-more Micro-Graphic EA1-T4CL 4-inch color touch panel, connected to the H2-DM1E CPU serial communication port (operator commands to the PLC; status/data back to the panel). Powered from the shared Mean Well NDR-480-24 24 VDC supply.
- **Safety circuit:** Hardwired E-stop loop and two-hand deadman start; motor contactor energized through the safety chain. Access guarding at the feed is handled by the ME team's mechanical design, so no separate electrical lid interlock is used.
- **24V DC supply:** Mean Well NDR-480-24 (480 W) for control voltage.

4.2 Signal Flow

- Operator inputs (deadman buttons, E-stop, reset) $\rightarrow$ PLC discrete inputs (D2-08ND3, sourcing mode).
- PLC outputs (D2-08TR relay module, common to PSU $-V$) $\rightarrow$ VFD FOR (Y7), REV (Y6), RST (Y5) inputs in sinking (NPN) mode.
- VFD fault relay (FC + FA, NO active-high) $\rightarrow$ PLC X5 input. X5 = HIGH when VFD trips on fault, LOW when healthy.
- HMI $\leftrightarrow$ PLC via serial (H2-DM1E CPU serial COM port; matched baud, parity, and stop bits on both ends).

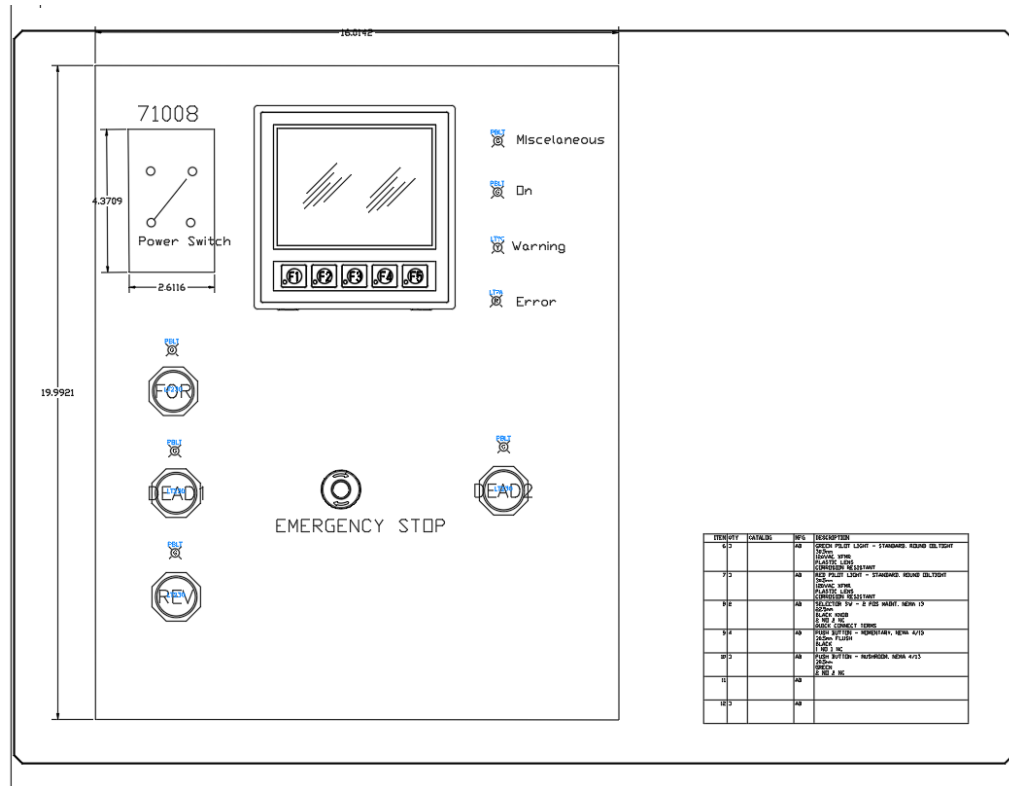


Figure 1: Operator control panel layout: HMI, power switch, forward/reverse and deadman pushbuttons, emergency stop, and status indicators.

[TODO: Add a system block diagram (power + signal flow) to complement the panel layout above.]

5 Hardware Design

5.1 Power Distribution

- Main disconnect / power switch: PowerTec 71008 two-pole (DPST) START/STOP switch, 120–230 VAC single-phase, rated 20 A / 3 HP max; switches both the line and neutral conductors.
- VFD branch breaker: 30 A C-curve DIN-rail (handles inrush, sized per NEC 430.52).
- 24V PSU branch breaker: 10 A C-curve DIN-rail.
- **Conductors:** power runs use 12 AWG copper (THHN/MTW), rated 20 A / 25 A / 30 A at the 60/75/90°C ampacity columns of NEC Table 310.16 – well above the motor’s 7.6 A FLA and the VFD input current. Control wiring uses 16–18 AWG,

landed directly as straight stripped conductors at the terminal blocks (no ferrules used in this build).

- **Fuse / OCPD sizing:** branch and component fuses are sized at **125% of the protected component's rated continuous load** per NEC 210.20(A) / 215.3. (The motor branch breaker is the exception – NEC 430.52 permits a higher multiple of FLC to ride through inrush; running overload is handled by the VFD's electronic motor-overload function.)
- All branches fused individually at terminal blocks for downstream loads.

5.2 Motor and VFD

The Huanyang HY02D211B-T was selected over alternatives via a decision matrix (price, documentation, reliability). A brake resistor on the P/Pr terminals absorbs the regenerative energy from rapid reversal during the jam-clear cycle. The drive is programmed as shown in Table 1.

Table 1: Programmed VFD parameters (Huanyang HY02D211B-T).

Param	What it does	Value	Why this value
PD141	Motor rated voltage	230 V	Matches the GE test-motor nameplate (230 V low-voltage connection); sets the base of the V/Hz curve.
PD142	Motor rated current (FLA)	7.6 A	Nameplate full-load amps; sets the drive's current limit and electronic-overload reference.
PD143	Number of motor poles	4	Nameplate; 4-pole = 1800 RPM synchronous, so the drive resolves speed and slip correctly.
PD144	Motor rated speed	1750 RPM	Nameplate full-load speed; with the pole count it lets the drive estimate slip.
PD002	Run / frequency source	1 (external)	Speed comes from the external 0–10 V analog input (VI) driven by the PLC, not the keypad.
PD052	Relay output (FA/FC) function	02 (fault)	Closes the FA–FC relay on any fault so the PLC reads a jam/trip on input X5.

continued on next page...

Param	What it does	Value	Why this value
PD054	Analog output (VO) function	1 (current)	Outputs 0–10 V proportional to motor current to the PLC analog input for monitoring.
PD123	Over-torque action	3 (trip)	Over-torque self-trips the drive (which fires the fault relay); see the note below.
PD124	Over-torque threshold	150%	Set between no-load current (~ 0.5 A) and measured jam current (> 11 A) so normal shredding does not trip but a jam does.
PD125	Over-torque detection time	3.0 s	The over-torque must persist 3 s before tripping, so brief load spikes (feeding a piece) do not nuisance-trip.

[NOTE: The over-torque relay function ($PD052 = 12$) did not fire the relay in our specific Huanyang revision when $PD123 = 2$ (continue running). Documented as a firmware-specific limitation (see Table 1). Working configuration uses $PD052 = 02$ (fault indication) and $PD123 = 3$ (over-torque triggers self-trip, which fires the fault relay). The PLC then runs the unjam routine including a RST pulse to clear the VFD’s latched fault state between retry attempts.]

5.3 Safety Circuit

- E-stop: NC contact (E22B1) hardwired in series with the SD-N35 contactor coil. Pressing it drops the coil and opens the contactor, removing motor power independently of the PLC.
- Two-hand deadman: both buttons must be held continuously to enable the motor (evaluated by the PLC; see the Safety Functions assessment below).
- Access guarding: handled by the ME team’s mechanical design (feed geometry), so no separate electrical lid interlock is used (this was a may-have, not a requirement).
- Motor contactor: the SD-N35 is energized through the hardwired safety chain, and the VFD is powered through it, so the safety chain has hardware precedence over the PLC.

[TODO: Add safety circuit schematic. Reference NFPA 79 section 9 for safety-related stop functions.]

5.4 Jam Detection Signal Chain

- VFD's multi-function relay (terminals FA, FB, FC) programmed for Fault Indication (PD052 = 02). **Only FA and FC are used** for jam detection; the FB (normally-closed) contact is left unconnected.
- Wired as an NO active-high pair: +24V (through a 500 mA fast-blow fuse) feeds the FA contact; the FC common returns to PLC X5 input on D2-08ND3 (Slot 2). When the VFD trips, FA–FC closes and +24V reaches X5.
- PLC sees X5 = LOW during normal operation, X5 = HIGH when VFD trips on any fault (over-torque, overvoltage, overheat, etc.).
- PLC ladder reads [X5] on rising edge to detect new fault events.
- Wire-break failure mode is handled at the safety-circuit level by the independent hard-wired E-stop loop and motor contactor, rather than at the jam-signal level.
- Full terminal-by-terminal wiring is given in the I/O point list (Appendix B) and the VFD parameter/wiring table (Appendix D).

6 Software Design

6.1 PLC Ladder Logic Architecture

- Do-more Designer stage program (similar to grafcet / SFC).
- States: IDLE, RUNNING, JAM_DETECTED, RST_PULSE, REVERSE, RESUME, LOCKOUT.
- Transitions driven by deadman input, X5 jam signal, jam counter, and reset button.

6.2 State Machine

- **IDLE** → **RUNNING** when both deadman buttons held AND not locked out AND startup mask done.
- **RUNNING** → **JAM_DETECTED** on rising edge of X5 (VFD trip → fault relay fires → X5 goes HIGH).
- **JAM_DETECTED** → **RST_PULSE** if jam count < 3, → **LOCKOUT** if jam count ≥ 3 .
- **RST_PULSE** (500 ms) → **REVERSE** (2 s).

- **REVERSE** → RESUME (brief delay) → back to RUNNING.
- **LOCKOUT** → IDLE only via dedicated reset button.

6.3 Jam Counter Reset Logic

- Counter increments on each X5 rising edge (each new VFD fault triggers a jam event).
- Resets to 0 if motor runs cleanly for 30 continuous seconds in the RUNNING state.
- Also resets if deadman is released for 30 continuous seconds (session reset).
- Whichever condition fires first wins.

6.4 Startup Mask

- For first 5 seconds after PLC power-up, X5 is ignored.
- In NO active-high wiring, this guards against transient X5 pulses during VFD initialization. Less critical than under NC wiring but kept as a defensive measure.

6.5 HMI Programming

- Status displays: Ready, Running, Jam in progress, Lockout, Fault.
- Operator inputs: Speed setpoint (currently fixed in PLC), Reset button.
- Data displayed: Motor current (from the VFD VO analog output into PLC input F2-08AD-1 CH1 – wired but not yet correctly scaled, see Open Issues), Jam count this session, Total runtime.

[QUESTION: We have not finalized the HMI screen layout. Should we include mock-ups in the rough draft, or wait for the final?]

7 Safety Design

7.1 NFPA 79 Compliance

- Wire color coding follows NFPA 79 §13.2: Black for AC line/load, Red for AC control, Blue for DC control, Blue/W stripe for DC return, Green/Yellow for ground.
- This color convention is applied to every conductor in the panel.
- All control wires labeled at both ends.

- Control wires land as straight stripped conductors at the terminal blocks (no ferrules used in this build).

7.2 Safety Functions

- **E-stop:** Category 0 stop (immediate removal of power). Hardwired — an E22B1 NC contact in series with the SD-N35 contactor coil; pressing it drops the coil and opens the contactor independently of the PLC. Single channel; the PLC additionally monitors the state on X4 (not a safety-rated channel).
- **Two-hand / deadman control:** both blue pushbuttons must be held continuously to enable the motor. As-built, the two NO inputs are AND-ed in the standard PLC's ladder logic to gate the VFD run output. It is the run-permissive and the safety interlock satisfying requirement M6, but it is *not* a safety-rated two-hand device per ISO 13851 Type III (no simultaneity timer, complementary contacts, or self-monitoring).
- **Access guarding:** handled by the ME team's mechanical design (feed geometry), so no electrical lid interlock is implemented (a lid interlock was a may-have, not a requirement).
- **Capacitive-touch module (planned):** a capacitive-touch safety module is on hand and the ladder logic to act on it (input X7 → stop motor + fault indication) is written, but it is *not yet wired*. This is a may-have (additional safety sensing) and is not relied upon in the current build.
- **Overload:** the VFD's electronic motor-overload function plus the over-torque trip (see Jam Detection); equipment protection rather than a personnel-safety function.

7.3 Performance Level (ISO 13849-1) — Preliminary Assessment

This is a preliminary, self-assessed determination from the as-built architecture. A formal verification needs a documented ISO 12100 risk assessment (to set the required performance level, PLr) and component reliability data (MTTFd / B10d, diagnostic coverage). Table 2 summarizes the achieved categories.

Table 2: Preliminary ISO 13849-1 assessment of the as-built safety functions.

Safety function	As-built architecture	Category (prelim.)
Emergency stop	Hardwired E22B1 NC contact in series with the SD-N35 contactor coil; opening it removes motor power. Single channel, well-tried components; PLC monitors X4 (not safety-rated).	Cat 1 (PL b–c)
Two-hand / deadman	Two NO buttons AND-ed in the standard PLC to gate the VFD run output. Single channel; no simultaneity check, complementary contacts, or self-monitoring.	Cat B (PL b); not ISO 13851 Type III
Over-torque / jam stop	VFD over-torque trip plus electronic motor overload. Equipment protection, not a personnel-safety function.	Not rated (process protection)

Overall the control system is approximately **Category B–1 (PL b–c)**, limited by: single-channel architecture throughout; a standard (non-safety) PLC evaluating the two-hand permissive; a single contactor with no force-guided (mechanically linked) contacts and no feed-back (EDM) monitoring to detect a welded contactor; and the as-built neutral-switching deviation (see the NEC note). The hardwired E-stop, built from well-tried components, is the strongest function (Category 1).

To raise the achieved PL — for example to Category 3 / PL d, should the risk assessment require it for the rotating-blade hazard — would need a dual-channel architecture, a safety relay or safety-rated PLC, two contactors (or force-guided contacts) with feedback monitoring, switching of the hot leg, and a synchronous two-hand module meeting ISO 13851.

[TODO: Confirm with the advisor whether formal PL/SIL verification is in cap-stone scope. If so, document the target PLr from the ISO 12100 risk assessment and the component MTTFd / B10d and DC values to complete the calculation.]

7.4 Failure Mode Analysis

- Wire break in jam signal: PLC sees X5 stuck LOW. Jam signal cannot fire, but the hardwired E-stop loop and motor contactor provide independent safety. Operator can also see motor stop and trigger investigation via fault-light absence.
- VFD loses power: the FA–FC fault relay de-energizes and opens, so X5 reads LOW (indistinguishable from healthy). The motor stops because the VFD itself is unpowered, so the condition is caught by the motor not running rather than by the jam signal.

- PLC fails: motor contactor drops out (safety chain bypasses PLC). Motor coasts to stop.
- Brake resistor disconnected: VFD will trip on overvoltage during deceleration. Motor coasts to stop. Diagnosable from VFD display.

8 Standards and Regulatory Compliance

This section consolidates every electrical and machine-safety standard the design is built against and states, for each, how it is applied to the shredder. Many of these are referenced in passing elsewhere in this report; they are gathered here so a reviewer (or a future maintenance team) can audit compliance in one place.

8.1 Standards Summary Matrix

Standard	Scope	How it is applied here
NFPA 70 (NEC)	National Electrical Code – premises wiring, motor circuits	Branch-circuit, conductor, grounding, and disconnect design (see §8.2).
NFPA 79	Electrical standard for industrial machinery	Wire color coding, supply disconnect, protective bonding, stop functions, operator devices (see §8.3).
NEMA 250	Enclosure type ratings	Control enclosure selected to a NEMA type matched to the lab/shop environment (see §8.4).
NEMA ICS 1/2	Industrial control & systems	Contactor / control device selection and marking.
NEMA MG 1	Motors and generators	Motor nameplate interpretation and VFD parameterization.
UL 508A	Industrial control panels	Panel construction practice and short-circuit current rating (SCCR) awareness.

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Standard	Scope	How it is applied here
IEC 60204-1	Safety of machinery – electrical equipment	International counterpart to NFPA 79; used to cross-check stop categories.
ISO 12100	Risk assessment / risk reduction	Framework behind the hazard analysis driving the safety circuit.
ISO 13849-1	Safety-related parts of control systems (PL)	Preliminary self-assessment (Cat B-1) of the E-stop and two-hand control chain.
ISO 13850	Emergency stop function	E-stop actuator type, color, and Category 0 behavior.
ISO 13851	Two-hand control devices	Deadman two-hand start timing and configuration.
OSHA 29 CFR 1910.147	Lockout/Tagout (LOTO)	Lockable main disconnect for energy isolation during service.

8.2 NFPA 70 – National Electrical Code

NEC article	Requirement	How it is applied here
Art. 430.6	Motor FLC basis	From the GE test-motor nameplate (FLA 7.6 A at 230 V); basis for all downstream sizing. Re-verified when the production motor is selected.
Art. 430.22	Branch conductor sizing	Conductors sized at $\geq 125\%$ of motor FLC.
Art. 430.32	Running overload	VFD electronic motor-overload (PD142 = 7.6 A) plus the over-torque trip.
Art. 430.52	Branch short-circuit / ground-fault	30 A C-curve breaker on the VFD branch; clears in-rush without nuisance trips (inverse-time-breaker allowance).
Art. 430.102/.109	Disconnecting means	Provided ahead of the controller and VFD.
Art. 430 Part X (430.122)	Adjustable-speed drive systems	Conductor and protection rules applied to the VFD-fed motor circuit.
Art. 250	Grounding & bonding	EGC (green / green-yellow) bonds the VFD, PSU, motor frame, and panel to a common ground bus and building earth.
Art. 310	Conductor ampacity	Conductors chosen from the ampacity tables for the installed temperature rating.
Art. 409	Industrial control panels	Panel layout, marking, and overcurrent considerations.
Art. 110.26	Working space	Clearances around live parts maintained for safe service. [TODO: Confirm final panel mounting leaves required clearance.]

[NOTE: As-built deviation flagged during the Point List review: the SD-N35 safety contactor switches the VFD neutral (T) leg rather than the hot (R) leg. NEC / NFPA 79 practice is to switch the ungrounded (hot) conductor so that opening the contactor de-energizes the load to ground potential. Switching the grounded conductor leaves the load at line potential when the contactor is open. This is documented in the as-built I/O point list (Appendix B,

VFD terminals) and must be corrected, or explicitly justified and risk-assessed, before final sign-off.]

8.3 NFPA 79 – Electrical Standard for Industrial Machinery

NFPA 79 clause	Requirement	How it is applied here
§5.3	Supply disconnecting means	Lockable main disconnect (PowerTec 71008) isolates the machine from supply; also satisfies LOTO.
§7	Overcurrent protection	Branch and downstream fusing at terminal blocks (PSU 25 A slow-blow, PLC 1 A fast-blow, control loads individually fused).
§8	Protective bonding circuit	Continuous equipment-grounding / bonding path to all exposed conductive parts.
§9	Control circuits & functions	Stop functions classified by stop category; the hard-wired safety chain has precedence over the PLC.
§10.7–10.8	Operator interface / E-stop	Red mushroom actuator on a yellow background, self-latching NC contact; start devices guarded by the two-hand requirement.
§13.2	Conductor color coding	Black AC line/load, red AC control, blue DC control, blue/white DC return, green/yellow PE; wires labeled both ends, landed straight (no ferrules in this build).

8.4 NEMA Standards

Standard	Covers	How it is applied here
NEMA 250	Enclosure type ratings	Targeting Type 12 (dust-tight, drip-tight) for the PLC/HMI panel. [TODO: Confirm final enclosure type and dust-ingress handling.]
NEMA ICS 1 / ICS 2	Industrial control & systems	Contactor and control-device selection/markings; the Mitsubishi SD-N35 is sized above motor FLA for AC-3 (motor) duty.
NEMA MG 1	Motors & generators	Interpreting the motor nameplate (voltage, poles, RPM, service factor) for VFD parameters PD141--PD144.

8.5 Machine-Safety Standards (ISO / IEC)

Standard	Covers	How it is applied here
ISO 12100	Risk assessment / reduction	Identifies the primary hazards (rotating blades, entanglement, electrical) and drives the layered safeguards.
ISO 13849-1	Safety-related control parts (PL)	Preliminary assessment: hardwired E-stop $\approx$ Cat 1; two-hand permissive $\approx$ Cat B (see the Safety Functions section).
ISO 13850	Emergency stop function	Category 0 stop (immediate removal of power) with fail-safe NC wiring.
ISO 13851	Two-hand control devices	Both buttons actuated within 0.5 s of each other and held continuously to enable the motor.
IEC 60204-1	Electrical equipment of machines	International analogue of NFPA 79; used to cross-check stop categories and supply disconnection.

8.6 Occupational Safety (OSHA)

Regulation	Covers	How it is applied here
29 CFR 1910.147	Lockout/Tagout (LOTO)	The main supply disconnect is lockable for positive isolation during blade-area service and maintenance.
29 CFR 1910 Subpart S	Electrical safety	General electrical-safety practices (guarding of live parts, grounding) observed throughout the panel build.

[TODO: Before final submission: convert any remaining “targets / intends to” language above into verified compliance statements, attach the panel enclosure datasheet, and resolve the NFPA 79 neutral-switching deviation note.]

9 Testing and Validation

9.1 Signal Chain Tests

- Manual jumper test (FA-FC short): confirmed PLC X5 reads LOW correctly – wiring intact.
- Guaranteed-trip test (PD124 = 1, PD125 = 5.0): VFD trips reliably on motor run, fault relay fires, PLC X5 drops.
- Threshold tuning: 150% (PD124 = 150) chosen for production based on no-load current (0.5 A) versus expected jam current (above 11 A actual).

9.2 Load Testing

[TODO: Schedule load testing with actual plastic samples once mechanical integration complete. Need ME team coordination.]

9.3 Safety Validation

- E-stop response time: [TODO: measure with scope].
- Two-hand control timing: [TODO: verify both buttons must be held continuously; simultaneous release stops within X ms].

10 Results

10.1 What Works

- All motor parameters programmed and verified.
- Over-torque detection chain validated (VFD detects, relay fires, PLC reads, ladder reacts).
- Fail-safe NC wiring confirmed – system correctly enters fault state on signal chain failure.
- HMI communicates with the PLC over the serial COM link.

10.2 Open Issues

- Vendor-specific Huanyang firmware quirk: PD052 = 12 does not fire relay in continue-running mode. Working around with PD052 = 02 + PD123 = 3.
- PLC ladder logic not fully complete – state machine implemented, but unjam routine timing needs tuning with real loaded motor.
- Mechanical integration pending – have not yet tested under real shredding load.
- Motor-current monitor (VFD V0 → F2-08AD-1 CH1) is wired, but under time constraints it was never configured/scaled correctly to read the VFD output current. The signal path exists; the VFD analog-output and PLC analog-input scaling still need to be set and calibrated.

[QUESTION: How much detail should we include about the Huanyang firmware quirk? It is an interesting design lesson (vendor lock-in, never trust the datasheet completely) but distracts from the high-level results.]

11 Discussion

11.1 Design Tradeoffs

- Cost versus reliability: chose Huanyang VFD (\$150) over ATO (\$450). Firmware quirks documented in this report (the Software Design note and Appendix D) for future maintenance.

- 110V single-phase versus 220V split-phase mains: 110V chosen for receptacle availability in PSU lab; trade-off is double the input current and need for a dedicated 30 A circuit.
- NO active-high versus NC fail-safe jam signal: the VFD fault relay is wired NO active-high (FA–FC closes on trip, X5 HIGH). Simpler to wire, but not fail-safe at the signal level—a broken wire reads as “no fault.” That residual risk is covered by the independent hardwired E-stop loop and motor contactor, which remove power regardless of the PLC.

11.2 Lessons Learned

- Always verify VFD relay function with multimeter before assuming firmware works as documented.
- Document deviations from datasheet behavior in version control so future teams can find them.
- Power cycle after parameter changes – some VFDs require it for save to fully take effect.
- Use ferrules on control wires in future builds: this build landed straight stranded wire at the terminal blocks, and loose/spread strands caused multiple debugging hours during commissioning.

11.3 Project Management Notes

[TODO: Add a paragraph on how the team coordinated, what tools we used (GitHub, weekly meetings, etc.), and how schedule slipped or held.]

12 Future Work and Improvements

12.1 Safety and Compliance

- **Integrate the capacitive-touch safety (CTSI) module:** wire the on-hand capacitive-touch module to input X7. The ladder logic to stop the motor and raise a fault on a touch trip is already written, so this is primarily a wiring and validation task. It adds a no-contact operator-presence safeguard at the feed.
- **Raise the safety performance level:** move from the current single-channel Cat B–1 toward Cat 3 / PL d — a safety relay or safety-rated PLC, dual-channel inputs,

two contactors (or force-guided contacts) with feedback (EDM) monitoring, and a synchronous two-hand module per ISO 13851.

- **Correct the supply switching:** switch the hot (ungrounded) conductor through the contactor rather than the neutral, per NFPA 79 / NEC.
- **Ferrule the control wiring:** crimp ferrules on stranded control conductors at the terminal blocks to eliminate the loose-strand faults seen during commissioning.

12.2 Monitoring and Connectivity

- **Telemetry:** stream live operating data (motor current, speed, jam count, fault state, runtime) off the PLC for dashboards and trend analysis.
- **Modbus to the VFD:** the Huanyang supports RS-485 Modbus (currently unused). Adding it would pull far richer data straight from the drive — output current, voltage, frequency/RPM, DC-bus voltage, drive temperature, and detailed fault codes — beyond the single 0–10 V analog monitor.
- **Internet / remote viewing:** expose a read-only view of device status over the network — via the H2-DM1E’s built-in web server or an ESP32 / edge gateway — so the shredder can be monitored remotely from a browser or phone.
- **Data logging:** record total runtime, jam events per shift, and faults locally for predictive maintenance.

12.3 Capability

- **Predictive jam detection:** train a small model on the motor-current waveform shape just before a jam to detect and pre-empt jams earlier.
- **Mechanical integration:** add a feed hopper that auto-meters plastic into the blades.
- **Higher-capacity model:** scale to a 5–10 HP drivetrain for greater throughput.

13 Ethics and Design Considerations

A summary of the ethics and design evaluation follows:

- **Public health and safety:** Yes – layered safety functions (hardwired E-stop, two-hand deadman control), NFPA 70 / NEMA compliance, hardwired safety chain.
- **Environmental:** Yes – core project purpose is diverting plastic from landfill.

- **Economic:** Yes – significant cost savings versus commercial alternatives, careful component selection within budget.
- **Global:** Possibly – approach is replicable globally, components internationally sourced.
- **Cultural:** No – industrial machine for B2B use, no culture-specific design.

14 References

- NFPA 70 (2023): National Electrical Code (esp. Articles 110, 250, 310, 409, 430).
- NFPA 79 (2024): Electrical Standard for Industrial Machinery.
- NEMA 250: Enclosures for Electrical Equipment.
- NEMA ICS 1 / ICS 2: Industrial Control and Systems.
- NEMA MG 1: Motors and Generators.
- UL 508A: Industrial Control Panels.
- IEC 60204-1:2016: Safety of Machinery – Electrical Equipment of Machines.
- ISO 12100:2010: Safety of Machinery – Risk Assessment and Risk Reduction.
- ISO 13849-1:2023: Safety-related parts of control systems.
- ISO 13850:2015: Emergency Stop Function.
- ISO 13851:2019: Two-hand control devices.
- OSHA 29 CFR 1910.147: The Control of Hazardous Energy (Lockout/Tagout).
- Huanyang HY-series VFD User Manual.
- AutomationDirect DL205 / Do-more User Manual.
- GE 5KE Frame 182T Motor Datasheet.

[TODO: Add full reference list with URLs / publication info before final.]

15 Appendices

15.1 Appendix A: Bill of Materials

Major purchased items for the electrical system (procurement BOM, rev 02/20/26):

Item	Description	Mfr. P/N	Critical
Motor	AC induction motor, 3-phase (bench-tested with GE 5KE182BC205B)	TBD	Yes
VFD	Huanyang 1-phase in / 3-phase out drive, 3 HP	HY02D211B-T	Yes
HMI	C-More Micro-Graphic 4” color touch panel	EA1-T4CL	Yes
Circuit breaker	DIN-rail branch breaker	NBE7	No
Power switch	Single-phase ON/OFF disconnect, 120–230 V	71008	No
Status lights	Red / green panel indicators	Ichor-provided	No
E-stop	Mushroom emergency-stop pushbutton	Ichor-provided	Yes

[NOTE: The final production motor is TBD (3–5 HP per requirements). Bench testing and the VFD motor parameters (PD141–144) use the GE 5KE182BC205B nameplate the team has on hand (3 HP, 230 V, 7.6 A, 4-pole, 1750 RPM); the production motor’s nameplate will supersede those values once selected. Additional control-panel hardware (PLC rack/modules, Mean Well NDR-480-24 PSU, SD-N35 contactor, Phoenix Contact breaker, brake resistor, illuminated pushbuttons, fuses) is itemized in the component tables of the Hardware Design section and the point list in Appendix B.]

15.2 Appendix B: PLC I/O Point List

The complete as-built point list, transcribed from the project workbook (`Point List.xlsx`), follows on the landscape pages below — every sheet, using the spreadsheet’s columns *Point Description*, *Origin Address*, *DO / DI / AO / AI / Pwr*, *Destination Address*, *Destination Description*, and *Notes*: the index; the Mitsubishi SD-N35 contactor; the Mean Well NDR-480-24 power supply; the DirectLOGIC 205 rack and the H2-DM1E CPU; the EA1-T4CL HMI; every I/O module (D2-08ND3, F2-08AD-1, F2-04RTD, D2-08TR, F2-08DA-2), the legacy D2-08TA and F2-02DAS-2 sheets; and the Huanyang HY02D211B-T VFD.

Sheet: Index

POINT INDEX	ASSIGNMENT					
		PAGE NO.	NO. OF SHEETS	DESCRIPTION		
		1	1	Mitsubishi SD-N35		
		2	2	Mean Well NDR-480-24		
		3	3	Direct Logic 205 D2-09B-1		
		REV	DATE	DESCRIPTION	Created By	Checked by
		1				
		2				
		2				
		2				
		3				
		4				

Sheet: Mitsubishi SD-N35

Point	Descrip- tion	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Descrip- tion	Notes
Contact	(Mit- subishi SD-N35) – Coil A1	A1	0	0	0	0	1	Y0	D2-08TD2 Output Mod- ule (12–24VDC)	A1 $\rightarrow$ DF103V (1/2A slow blow fuse) $\rightarrow$ D2-08TD2 output supply- ing +24VDC
Contact	(Mit- subishi SD-N35) – Coil A2	A2	0	0	0	0	1	-V	NDR-480-24 Power Sup- ply	A2 $\rightarrow$ direct return to power sup- ply negative (–V)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Contactor (Mitsubishi SD-N35) – Line Input	L1	0	0	0	0	1	L	Phoenix Contact TMC 83C 15A (2907618)	L1 $\rightarrow$ Phoenix Contact fuse breaker $\rightarrow$ fed from wall AC line
Contactor (Mitsubishi SD-N35) – Load Output	T1	0	0	0	0	1	R	HY Series VFD (HY02D211B-T)	T1 $\rightarrow$ passes through contactor $\rightarrow$ supplies VFD terminal R (line input)
	Total Points								

Sheet: Mean Well NDR-480-24

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
		0					AC Mains Hot from wall via main power switch and 25 A fuse	120 VAC line conductor from wall outlet, routed through panel main disconnect, then through 25 A slow-blow fuse on EURO S10H-5H fuse block	Wall AC L $\rightarrow$ main disconnect $\rightarrow$ EURO S10H-5H 25 A slow-blow $\rightarrow$ PSU L. Black wire per NFPA 79.
Mean Well NDR-480-24 N (AC Neutral Input)	N	0	0	0	0	1		120 VAC neutral conductor from wall outlet, returns through neutral bus	Wall AC N $\rightarrow$ neutral bus $\rightarrow$ PSU N. White wire per NFPA 79.
Mean Well NDR-480-24 G (Ground / PE)	G	0	0	0	0	1	Panel Ground Bus	Equipment grounding conductor; bonds PSU chassis to panel ground bus and ultimately to building earth	PSU G stud $\rightarrow$ panel ground bus. Green or Green/Yellow wire per NFPA 79.

Point	Descrip- tion	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Mean Well NDR-480-24 +V to EURO S4LH (Pushbutton Common)		+V	0	0	0	0	1	EURO S4LH Terminal Block -> Pushbutton Commons	Fused +24V distribution to operator pushbutton common via S4LH	+V -> 1 A fast-blow fuse (EURO S4LH terminal) -> distributes to common side of all illuminated pushbuttons (Forward S2PR-P3G, Reverse S2PR-P3Y, Run 1/2 S2PR-P3B x2, Reset S2PR-P3R).
Mean Well NDR-480-24 +V to E22B1 E-Stop Circuit		+V	0	0	0	0	1	Cutler-Hammer E22B1 NC contact + SD-N35 contactor coil A1	Direct +24V (no fuse) into E-stop safety chain	+V -> E22B1 NC contact -> SD-N35 A1 (contactor coil). PLC X4 taps this line between E22B1 and A1 to monitor E-stop state.
Mean Well NDR-480-24 +V to VFD FA (Jam Signal)		+V	0	0	0	0	1	VFD Huanyang HY02D211B-T FA terminal	Direct +24V (no fuse) into VFD jam signal circuit	+V -> VFD FA. When VFD fault relay closes (PD052=02), FA-FC bridges and delivers +24V to PLC X5.
Mean Well NDR-480-24 +V to F2-08AD-1 (Module Power)		+V	0	0	0	0	1	F2-08AD-1 Analog Input Module +24V terminal	Direct module power	+V -> F2-08AD-1 +24V terminal. No fuse on this branch.
Mean Well NDR-480-24 +V to F2-08DA-2 (Module Power)		+V	0	0	0	0	1	F2-08DA-2 Analog Output Module +24V terminal	Direct module power	+V -> F2-08DA-2 +24V terminal. No fuse on this branch.
Mean Well NDR-480-24 +V to F2-04RTD (Module Power)		+V	0	0	0	0	1	F2-04RTD RTD Input Module +24V terminal	Direct module power	+V -> F2-04RTD +24V terminal. No fuse on this branch.

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Mean Well NDR-480-24	+V to Indicator LEDs	+V	0	0	0	0	1	LED indicator (+) terminals (SA-LDG/Y/B/B/R, Wamco 525)	Direct +24V to (+) side of all illuminated push-button LED blocks and the separate Wamco error light	+V -> SA-LDG (+), SA-LDY (+), SA-LDB (+) x2, SA-LDR (+) (when wired), Wamco 525 (+). LED (-) sides return to D2-08TR relay outputs Y0-Y4 which sink to -V when energized.
Mean Well NDR-480-24	-V to SD-N35 Contactor Coil A2	-V	0	0	0	0	1	Mitsubishi SD-N35 Contactor A2 (coil return)	Contactor coil return through E-stop chain	-V -> SD-N35 A2 terminal. Coil energizes when E22B1 NC closes (E-stop not pressed) and +V reaches A1.
Mean Well NDR-480-24	-V to D2-08ND3 (Input Common)	-V	0	0	0	0	1	D2-08ND3 Discrete Input Module Common (C)	Module input common for sourcing mode operation	-V -> D2-08ND3 C terminal. Sourcing mode reference - X inputs read HIGH when +24V is applied.
Mean Well NDR-480-24	-V to D2-08TR (Relay Common)	-V	0	0	0	0	1	D2-08TR Relay Output Module Common (C)	Relay common for sinking mode operation	-V -> D2-08TR C terminal. When any Y0-Y7 relay closes, that output terminal connects to -V (sinks current from the load).
Mean Well NDR-480-24	-V to F2-08AD-1 (Module Power Return)	-V	0	0	0	0	1	F2-08AD-1 Analog Input Module 0V terminal	Module power return and analog signal common	-V -> F2-08AD-1 0V terminal.
Mean Well NDR-480-24	-V to F2-08DA-2 (Module Power Return / Analog Common)	-V	0	0	0	0	1	F2-08DA-2 Analog Output Module 0V terminal	Module power return and analog signal reference for all 8 channels	-V -> F2-08DA-2 0V terminal. Single 0V terminal also serves as the analog signal common.

Point	Descrip- tion	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Descrip- tion	Notes
Mean Well NDR-480-24	-V to F2-04RTD (Module Power Return)	-V	0	0	0	0	1	F2-04RTD RTD Input Module 0V terminal	Module power return	-V -> F2-04RTD 0V terminal.
Mean Well NDR-480-24	-V to VFD DCM (Digital Common)	-V	0	0	0	0	1	VFD Huanyang HY02D211B-T DCM terminal	VFD digital input common reference	-V -> VFD DCM. Provides 0V reference for FOR/REV/RST inputs (sinking/NPN mode). Closes the loop when D2-08TR relays Y5/Y6/Y7 pull those VFD inputs to DCM.
Mean Well NDR-480-24	-V to VFD ACM (Analog Common)	-V	0	0	0	0	1	VFD Huanyang HY02D211B-T ACM terminal	VFD analog signal common reference	-V -> VFD ACM. Shared 0V reference for the analog speed command (VI from F2-08DA-2) and the analog current monitor (VO to F2-08AD-1).
Mean Well NDR-480-24	+V to HMI (EA1-T4CL)	+V	0	0	0	0	1	C-More EA1-T4CL HMI +VDC input	HMI panel power	+V -> HMI +VDC. 24 VDC supply for the C-More Micro-Graphic panel (replaces former Rhino PSB12-030-P).
Mean Well NDR-480-24	-V to HMI (EA1-T4CL)	-V	0	0	0	0	1	C-More EA1-T4CL HMI 0VDC return	HMI panel power return	-V -> HMI 0VDC. System DC common reference for the HMI panel.
Total Points										

Sheet: Direct Logic 205 D2-09B-1

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
PLC Rack (Direct-LOGIC 205) – Base		Base	0	0	0	0	1	D2-09B-1	Base / Power Unit	Rack base unit providing backplane power and I/O bus to all installed modules. Supports up to 9 I/O slots.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
PLC Rack (Direct-LOGIC 205) – Line Input	L	0	0	0	0	1	L	MORSETTITALIA EURO S10H-5H Fuse Block	Line from wall -> passes through power switch (closed = line passes) -> enters EURO S10H-5H fuse block with 1A fast blow fuse (PLC branch) -> enters PLC L terminal
PLC Rack (Direct-LOGIC 205) – Neutral Input	N	0	0	0	0	1	N	Neutral Bus	Neutral from wall -> directly connected to PLC N terminal
PLC Rack (Direct-LOGIC 205) – Ground (Protective Earth)	G	0	0	0	0	1	G	Ground Bus	Protective earth -> connected to PLC ground terminal / chassis bond
PLC Rack (Direct-LOGIC 205) – Slot 1	Slot 1	0	0	0	0	0	H2-DM1E	CPU / Ethernet Module	CPU module with built-in Ethernet port. Executes the ladder logic program and handles network communications.
PLC Rack (Direct-LOGIC 205) – Slot 2	Slot 2	0	8	0	0	0	D2-08ND3	8-Point Discrete Input Module	8-channel 24VDC discrete (digital) input module. Accepts sink/source wiring. Used for reading on/off signals such as pushbuttons, limit switches, and proximity sensors.
PLC Rack (Direct-LOGIC 205) – Slot 3	Slot 3	0	0	0	8	0	F2-08AD-1	8-Channel Analog Current Input Module (4-20 mA)	8-channel current input module. Used for motor current monitoring (CH1+ to VFD VO terminal, scaled 0-10V from VFD).
PLC Rack (Direct-LOGIC 205) – Slot 4	Slot 4	0	0	0	4	0	F2-04RTD	4-Channel RTD Input Module	4-channel RTD (Resistance Temperature Detector) input module. Compatible with PT100/PT1000 and other RTD types. Used for precision temperature measurement.

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
PLC Rack (Direct-LOGIC 205) – Slot 5		Slot 5	8	0	0	0	0	D2-08TR	8-Point Relay Output Module	8-channel relay (dry contact) output module. Common to PSU -V (sinking mode). Drives indicator LEDs (Y0-Y4) and VFD digital inputs (Y5=RST, Y6=FOR, Y7=REV).
PLC Rack (Direct-LOGIC 205) – Slot 6		Slot 6	0	0	8	0	0	F2-08DA-2	8-Channel Analog Voltage Output Module (0-10 V)	8-channel voltage output module. CH1 (+V1) provides VFD speed reference to VI terminal (0-10V scaled to frequency).
PLC Rack (Direct-LOGIC 205) – Slot 7		Slot 7	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.
PLC Rack (Direct-LOGIC 205) – Slot 8		Slot 8	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.
PLC Rack (Direct-LOGIC 205) – Slot 9		Slot 9	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.

Sheet: DirectLOGIC CPU (H2-DM1E)

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Power and communication interface	Backplane Connector (Left Edge)	0	0	0	0	0	1	D2-09B-1 Base Slot 1 Connector	Provides power and backplane communication	—

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Ground reference		Backplane GND	0	0	0	0	1	D2-09B-1 Base Ground Bus	Common ground for logic power	–
HMI communication port		CPU Comm Port	0	0	0	0	0	HMI EA1-T4CL COM Port	4” C-More Micro-Graphic panel (operator interface)	CPU communication port links to the C-More HMI (see HMI sheet). Match baud, parity, and stop bits on both ends. Carries operator commands to the PLC and status/data back to the panel.
		Total Points								

Sheet: Output Module (D2-08TA)

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Output Common (Hot)		C	1	0	0	0	1	120 V Hot Bus	ZipLink Terminal Block (ZL-RTB20-OUT). 1A in-line slow blow fuse	120 V hot feed shared by all outputs
Output 0		OUT0	1	0	0	0	0	A1	TX Contactor (GH15-BN)	Drives TX coil via ZipLink
Output 1		OUT1	1	0	0	0	0	A1	BUS Contactor (GH15-BN)	Drives BUS coil via ZipLink
Output 2		OUT2	1	0	0	0	0	A1	F1 Contactor (GH15-BN)	Drives F1 coil via ZipLink
Output 3		OUT3	1	0	0	0	0	95	Thermal Overload Relay (RTD32-180)	Feeds NC contact 95 $\rightarrow$ 96 $\rightarrow$ F2 coil
Module Power Feed		Backplane	0	0	0	0	1	Slot 3 Connector	PLC base backplane (120 VAC)	No external wiring

Point	Descrip- tion	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination	Descrip- tion	Notes
		Total Points									

Sheet: HMI (EA1-T4CL)

Point	Descrip- tion	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination	Descrip- tion	Notes
HMI Serial Comm Port		COM Port	0	0	0	0	0	Serial Port	H2-DM1E		Serial link between PLC CPU and C-More HMI. Configure PLC and HMI ports for matching baud, parity, stop bits
HMI Power +V		+VDC	0	0	0	0	1	+V	Mean Well NDR-480-24		HMI 24 VDC input powered from Mean Well NDR-480-24 +V rail
HMI Power 0 V		0VDC	0	0	0	0	1	0V	Mean Well NDR-480-24		0 VDC return to Mean Well NDR-480-24 -V rail (system DC common)
HMI Ground		Ground	0	0	0	0	1	Ground	Ground		
		Total Points									

Sheet: Output Module (F2-02DAS-2)

Point	Descrip- tion	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination	Descrip- tion	Notes
Analog (CH1)	Common	CH1-V	0	0	1	0	0	CM	VFD (GS1-20P2) through Ziplink ZL-RTB20		0V1 and CH1-V are internally shorted
Analog CH1	Output	CH1+V	0	0	1	0	0	AL	VFD (GS1-20P2) though Ziplink ZL-RTB20		0–10 V speed command to VFD

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Module Logic +24 V		+V1	0	0	0	0	1	VDC +24V	PLC Base (D2-09B-1) through ZipLink ZL-RTB20	Powers analog output circuitry, also fused with 1A fast blow fuse
Module Logic 0 V		0V1	0	0	0	0	1	VDC – 0V	PLC Base (D2-09B-1) through ZipLink ZL-RTB20	Common return for module logic
Slot Power Feed		Backplane 0	0	0	0	0	1	PLC Base Backplane	Internal slot power	No external wiring
		Total Points								

Sheet: Input Module (D2-08ND3)

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08ND3 X0		X0	0	1	0	0	0	Forward Push-button	Forward button S2PR-P3G with Contact block SA-CA with LED module SA-LDG	Shared button +24V -> NO contact -> X0.
D2-08ND3 X1		X1	0	1	0	0	0	Reverse Push-button	Reverse button S2PR-P3Y with Contact block SA-CA with LED module SA-LDY	Shared button +24V -> NO contact -> X1.
D2-08ND3 X2		X2	0	1	0	0	0	Run 1 / Deadman 1 Pushbutton	Run 1 button S2PR-P3B with Contact block SA-CA with LED module SA-LDB	Shared button +24V -> NO contact -> X2.

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08ND3 X3		X3	0	1	0	0	0	Run 2 / Dead-man 2 Pushbutton	Run 2 button S2PR-P3B with Contact block SA-CA with LED module SA-LDB	Shared button +24V -> NO contact -> X3.
D2-08ND3 X4		X4	0	1	0	0	0	E-Stop Button Monitor	E-stop button E22B1 with NC contact block, wired in series with Mitsubishi SD-N35 contactor coil	PSU +V -> E22B1 NC -> SD-N35 A1 (X4 tapped between E22B1 and A1). SD-N35 A2 -> PSU -V.
D2-08ND3 X5		X5	0	1	0	0	0	VFD Fault Relay	VFD fault output Huanyang HY02D211B-T internal SPDT relay (FA/FC/FB), programmed as fault indication via PD052=02	PSU +V -> VFD FA. VFD FC -> X5. FB unused.
D2-08ND3 X6		X6	0	1	0	0	0	Reserved - Reset Pushbutton (planned)	Reset button S2PR-P3R with Contact block SA-CA with LED module SA-LDR	Unwired. Planned: shared button +24V -> NO contact -> X6 (same topology as X0-X3).
D2-08ND3 X7		X7	0	1	0	0	0	Spare	Unused	Unwired.
D2-08ND3 Input Common (C)		C	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Both top-row and bottom-row C terminals on the D2-08ND3 are internally tied.	C -> PSU -V. Module wired in sourcing mode. Shared +24V button supply is fed from PSU +V through a 1 A fast-blow fuse (EURO S10H-5H holder) into the EURO S4LH distribution terminal, which provides the common +24V to X0/X1/X2/X3 (and X6 when wired).
Total Points										

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-08AD-1	+24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal directly. No fuse on this branch.
F2-08AD-1	0V (Module Power Return)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Shared with VFD DCM and ACM (all tied to PSU -V).	PSU -V -> Module 0V terminal. Also serves as the common reference for channel inputs.
F2-08AD-1	CH1+ (Channel 1 Signal)	CH1+	0	0	0	1	0	VFD Motor Current Monitor	VFD analog output Huanyang HY02D211B-T VO terminal, configured for output current via PD054=1 (0-10V proportional to motor current)	VFD VO -> Module CH1+ terminal. Signal return reference via PSU -V (VFD ACM also tied to PSU -V).
F2-08AD-1	CH2	CH2	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1	CH3	CH3	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1	CH4	CH4	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1	CH5	CH5	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1	CH6	CH6	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1	CH7	CH7	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1	CH8	CH8	0	0	0	1	0	Spare	Unused channel	Unwired.
Total Points										

Sheet: RTD Input (F2-04RTD)

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-04RTD	+24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal. Module installed in rack but no channels currently used in this build.
F2-04RTD	0V (Module Power Return)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail	PSU -V -> Module 0V terminal.
F2-04RTD	CH1	CH1	0	0	0	1	0	Spare	Unused RTD input channel	Unwired. Available for future RTD temperature sensor.
F2-04RTD	CH2	CH2	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
F2-04RTD	CH3	CH3	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
F2-04RTD	CH4	CH4	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
Total Points										

Sheet: Relay Output (D2-08TR)

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08TR	Y0	Y0	1	0	0	0	0	Forward Indicator LED	LED module SA-LDG (green, 12-24 VDC) integrated in Forward pushbutton S2PR-P3G (assembly also drives X0)	PSU +V -> SA-LDG (+). SA-LDG (-) -> Y0 contact. Relay sinks Y0 to -V when energized, lighting the LED.

Point	Descrip- tion	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Descrip- tion	Notes
D2-08TR	Y1	Y1	1	0	0	0	0	Reverse Indica- tor LED	LED module SA-LDY (yellow, 12-24 VDC) integrated in Reverse pushbutton S2PR-P3Y (assembly also drives X1)	PSU +V -> SA-LDY (+). SA-LDY (-) -> Y1 contact. Relay sinks Y1 to -V when energized.
D2-08TR	Y2	Y2	1	0	0	0	0	Run 1 / Dead- man 1 Indicator LED	LED module SA-LDB (blue, 12-24 VDC) integrated in Run 1 push- button S2PR-P3B #1 (assembly also drives X2)	PSU +V -> SA-LDB (+). SA-LDB (-) -> Y2 contact.
D2-08TR	Y3	Y3	1	0	0	0	0	Run 2 / Dead- man 2 Indicator LED	LED module SA-LDB (blue, 12-24 VDC) integrated in Run 2 push- button S2PR-P3B #2 (assembly also drives X3)	PSU +V -> SA-LDB (+). SA-LDB (-) -> Y3 contact.
D2-08TR	Y4	Y4	1	0	0	0	0	Error / Fault In- dicator	Wamco model 525 panel- mount LED indicator (28 VDC, 0.6 W)	PSU +V -> Wamco 525 (+). Wamco 525 (-) -> Y4 contact. 28V-rated indicator slightly under-driven at 24V (acceptable, ~21 mA draw).
D2-08TR	Y5	Y5	1	0	0	0	0	VFD Fault Re- set	Huanyang HY02D211B- T VFD RST terminal (multi-input, sink- ing/NPN mode)	Y5 -> VFD RST. When Y5 energized, relay closes Y5 to common (PSU -V), pulling RST input to DCM. ~500 ms pulse during PLC un- jam routine.
D2-08TR	Y6	Y6	1	0	0	0	0	VFD Forward Run	Huanyang HY02D211B- T VFD FOR termi- nal (multi-input, sink- ing/NPN mode)	Y6 -> VFD FOR. When Y6 energized, pulls FOR to DCM. Forward shredding.

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08TR	Y7	Y7	1	0	0	0	0	VFD Reverse Run	Huanyang HY02D211B-T VFD REV terminal (multi-input, sinking/NPN mode)	Y7 -> VFD REV. When Y7 energized, pulls REV to DCM. Used during PLC unjam routine.
D2-08TR	Relay Common (C)	C	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail	C -> PSU -V. Sinking mode: when any Y0-Y7 closes, that terminal connects to -V.
Total	Points									

Sheet: Analog Output (F2-08DA-2)

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-08DA-2	+24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal directly. No fuse on this branch.
F2-08DA-2	0V (Module Power Return / Analog Common)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Single 0V terminal serves both as power return and analog signal reference for all 8 channels.	PSU -V -> Module 0V terminal. VFD ACM is also tied to PSU -V, so the analog signal return is shared through the common 0V rail.
F2-08DA-2	+V1 (Channel 1 Signal)	+V1	0	0	1	0	0	VFD Analog Speed Reference	Huanyang HY02D211B-T VFD VI terminal (analog voltage input, 0-10 V), used as frequency/RPM reference. Requires PD002=1.	+V1 -> VFD VI. 0-10 V signal scaled to commanded frequency: 0 V = 0 Hz, 10 V = max frequency.

Point	Descrip- tion	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination	Descrip- tion	Notes
F2-08DA-2	+V2	+V2	0	0	1	0	0	Spare	Unused channel		Unwired.
F2-08DA-2	+V3	+V3	0	0	1	0	0	Spare	Unused channel		Unwired.
F2-08DA-2	+V4	+V4	0	0	1	0	0	Spare	Unused channel		Unwired.
F2-08DA-2	+V5	+V5	0	0	1	0	0	Spare	Unused channel		Unwired.
F2-08DA-2	+V6	+V6	0	0	1	0	0	Spare	Unused channel		Unwired.
F2-08DA-2	+V7	+V7	0	0	1	0	0	Spare	Unused channel		Unwired.
F2-08DA-2	+V8	+V8	0	0	1	0	0	Spare	Unused channel		Unwired.
Total Points											

Sheet: VFD (Huanyang HY02D211B-T)

Point	Descrip- tion	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination	Descrip- tion	Notes
VFD R	(AC Line Hot Input)	R	0	0	0	0	1	AC Mains Hot via main disconnect and 30 A breaker	120 VAC line conductor from wall outlet, routed through panel main disconnect and a 30 A C-curve DIN-rail circuit breaker		Wall AC L -> disconnect -> 30 A breaker -> VFD R. Black wire per NFPA 79.

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD T	(AC Neutral Input via Contactor and Breaker)	T	0	0	0	0	1	AC Mains Neutral via Phoenix Contact TMC 8 3C 15A breaker and Mitsubishi SD-N35 contactor	Neutral leg switched through the safety-disconnect contactor. As-built path: Wall N -> Phoenix Contact TMC 8 3C 15A breaker (P/N 2907618) -> SD-N35 T1 -> contactor pole -> SD-N35 L1 -> VFD T.	AS-BUILT: neutral leg is switched by the contactor instead of the hot leg. Note: NFPA 79 / NEC standard practice is to switch the HOT leg. Review for final report. When E-stop is pressed, SD-N35 coil drops -> contact opens -> VFD T loses neutral -> VFD shuts down.
VFD	Ground (PE)	Ground	0	0	0	0	1	Panel Ground Bus	Equipment grounding conductor; bonds VFD chassis to panel ground bus	VFD ground stud -> panel ground bus. Green or Green/Yellow wire.
VFD U	(Motor Phase 1 Output)	U	0	0	0	0	1	Banana Socket (Blue) -> Motor T1	Panel-mount banana socket (blue, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T1	VFD U -> blue banana socket -> motor terminal.
VFD V	(Motor Phase 2 Output)	V	0	0	0	0	1	Banana Socket (White) -> Motor T2	Panel-mount banana socket (white, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T2	VFD V -> white banana socket -> motor terminal.
VFD W	(Motor Phase 3 Output)	W	0	0	0	0	1	Banana Socket (Red) -> Motor T3	Panel-mount banana socket (red, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T3	VFD W -> red banana socket -> motor terminal.

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD P+ (Brake Resistor +)	P+		0	0	0	0	1	Brake Resistor ATO APCS-300R30-AD (+ terminal)	Dynamic braking resistor 300 W, 30 Ohms, aluminum-clad (ATO model APCS-300R30-AD)	VFD P+ -> brake resistor terminal 1.
VFD PR (Brake Resistor Return)	PR		0	0	0	0	1	Brake Resistor ATO APCS-300R30-AD (return terminal)	Dynamic braking resistor (same component as P+ row)	VFD PR -> brake resistor terminal 2.
VFD UPF	UPF		0	1	0	0	0	Unused	Multi-Output 2 (PD051), open-collector optocoupler	Unwired.
VFD DRV	DRV		0	1	0	0	0	Unused	Multi-Output 1 (PD050), open-collector optocoupler	Unwired.
VFD DCM (Digital Common, top row)	DCM		0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return - shared reference for FOR/REV/RST sinking inputs	VFD DCM (top row) -> PSU -V. PSU -V also feeds D2-08TR relay common.
VFD SPL	SPL		0	1	0	0	0	Unused	Multi-Input 6 - default low-speed select	Unwired.
VFD SPM	SPM		0	1	0	0	0	Unused	Multi-Input 5 - default middle-speed select	Unwired.
VFD SPH	SPH		0	1	0	0	0	Unused	Multi-Input 4 - default high-speed select	Unwired.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD RST (Fault Reset Input)	RST	0	1	0	0	0	D2-08TR Y5	PLC relay output (sinking NPN mode) driving fault reset	PLC Y5 -> VFD RST. Pulled to DCM (0V) when Y5 closes.
VFD REV (Reverse Run Input)	REV	0	1	0	0	0	D2-08TR Y7	PLC relay output (sinking NPN mode) driving reverse run command	PLC Y7 -> VFD REV. Pulled to DCM (0V) when Y7 closes.
VFD FOR (Forward Run Input)	FOR	0	1	0	0	0	D2-08TR Y6	PLC relay output (sinking NPN mode) driving forward run command	PLC Y6 -> VFD FOR. Pulled to DCM (0V) when Y6 closes.
VFD ACM (Analog Common, top row)	ACM	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return - shared reference for analog signals	VFD ACM (top row) -> PSU -V. Shared 0V reference for VI (speed in) and VO (current monitor out).
VFD VO (Analog Output, 0-10 V)	VO	0	0	1	0	0	F2-08AD-1 CH1+	PLC analog input module CH1, used as motor current monitor (PD054=1)	VFD VO -> F2-08AD-1 CH1+. 0-10 V scaled to motor output current.
VFD 10V	10V	0	0	1	0	0	Unused	+10 V reference output (for external potentiometer)	Unwired.
VFD FA (Multi-Output 3 Relay NO)	FA	0	1	0	0	0	Mean Well NDR-480-24 +V	DC power supply +V rail	PSU +V -> VFD FA. No fuse on this branch. FA-FC closes when VFD trips on any fault (PD052=02).
VFD FC (Multi-Output 3 Relay Common)	FC	0	1	0	0	0	D2-08ND3 X5	PLC discrete input X5 (jam signal)	VFD FC -> PLC X5. When FA-FC closes, +24V from FA reaches X5 (PLC reads HIGH = jam).
VFD FB (Multi-Output 3 Relay NC)	FB	0	1	0	0	0	Unused	Normally-closed contact - not used in active-high NO configuration	Unwired.

Point	Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD 24V	(On-board 24V Supply)	24V	0	0	0	0	1	Unused	Onboard 24 VDC auxiliary supply, max 200 mA	Unwired. Panel uses external Mean Well NDR-480-24 instead.
VFD DCM	(bottom row)	DCM (bottom)	0	0	0	0	1	(same node as top-row DCM)	Internally tied to top-row DCM	Alternate landing point. Top-row DCM is used.
VFD 5V		5V	0	0	1	0	0	Unused	+5 V reference output	Unwired.
VFD ACM	(bottom row)	ACM (bottom)	0	0	0	0	1	(same node as top-row ACM)	Internally tied to top-row ACM	Alternate landing point. Top-row ACM is used.
VFD AI	(Analog Current Input, 4-20 mA)	AI	0	0	0	1	0	Unused	Analog current input for speed reference. Speed command uses VI voltage input instead.	Unwired.
VFD VI	(Analog Voltage Input, 0-10 V)	VI	0	0	0	1	0	F2-08DA-2 +V1	PLC analog output module CH1, used as commanded frequency reference (PD002=1)	F2-08DA-2 +V1 -> VFD VI. 0-10 V scaled to commanded frequency.
VFD RS-		RS-	0	0	0	0	1	Unused	RS-485 Modbus communication negative line	Unwired. Modbus not used.
VFD RS+		RS+	0	0	0	0	1	Unused	RS-485 Modbus communication positive line	Unwired.
Total Points										

15.3 Appendix C: Wiring Diagrams

[TODO: Insert exported PDFs from AutoCAD: power, VFD/motor, PLC/HMI, power distribution.]

15.4 Appendix D: VFD Parameter Settings

The full programmed parameter set — code, function, value, and rationale — is given in Table 1 (Section 5.2, Motor and VFD). Motor values PD141–144 reflect the GE 5KE182BC205B test motor; the production motor is TBD and its nameplate will supersede these.

Power / braking wiring: R = AC hot (30 A breaker); T = AC neutral via SD-N35 contactor; U/V/W to motor via colored banana sockets; P+/PR to a 300 W, 30 Ω brake resistor (ATO APCS-300R30-AD) to absorb regenerative energy during reversal in the jam-clear cycle. Control wiring is summarized in Appendix B.

[NOTE: Firmware-specific finding: PD052 = 12 (continue-running over-torque indication) did not fire the relay on our Huanyang revision; the working configuration is PD052 = 02 with PD123 = 3 (over-torque self-trips, which fires the fault relay and is read by the PLC on X5).]

15.5 Appendix E: Test Plan

Test Plan v1.0 comprises a top-down power-on walkthrough followed by ten bottom-up test cases. Each case is exercised against the must/should requirements (M-series).

ID	Test case
Top-down	Power-on walkthrough: AC in → PSU → PLC boot → HMI IDLE/READY
BU-01	AC power input and fuse distribution
BU-02	24 VDC PSU output (Mean Well NDR-480-24)
BU-03	PLC rack initialization (DirectLOGIC 205)
BU-04	E-stop hardware safety circuit
BU-05	Two-hand start (deadman) safety logic
BU-06	Lid interlock (n/a in this build — access guarded mechanically by the ME team)
BU-07	VFD speed command and motor RPM (HY02D211B-T)
BU-08	Overcurrent / overload protection
BU-09	HMI operator interface (EA1-T4CL C-More Micro-Graphic)
BU-10	Status indicator lights (power / running / fault)

Validation results to date are summarized in the Testing and Validation and Results sections; load testing under real shredding load is pending mechanical integration.

16 Authors and Acknowledgements

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Disclaimer

This report documents an in-progress senior capstone design. Findings, parameters, and as-built details reflect the current state of the prototype and are subject to revision through final validation.

End of rough draft. Comments and questions welcome.
